

MDI3003 — Advanced Predictive Analytics

Medical Diagnosis Support: Disease Classification Using Decision Trees

Prepared by

Mohan Raj I

Register Number: 23MID0121

Course: MDI3003 — Advanced Predictive Analytics

Vellore Institute of Technology, Vellore

Fall Semester 2026–27

Item	Detail
Registration Number	23MID0121
Course	MDI3003 – Advanced Predictive Analytics
Faculty	Dr. Durgesh Kumar, Assistant Professor (Senior), SCOPE, VIT Vellore
Primary Method	Decision Tree Classification using CART
Model Progression	Basic CART → Tuned & Pruned CART → Random Forest
Core Dataset	Breast Cancer Wisconsin (Diagnostic), n = 569
Extended Dataset	UCI Heart Disease (Cleveland), n = 303
Environment	Python 3.13, scikit-learn (Jupyter Notebook)
Random Seed	42 (5-fold stratified cross-validation)

Table of Contents

1. Lab Title and Positioning

2. Lab Objectives
3. Learning Outcomes
4. Problem Statement
5. Real-World Motivation and Industry Relevance
6. Scope, Schedule, and Deliverables
7. Required Software and Python Libraries
8. Recommended Real-World Datasets
9. Theoretical Background
10. End-to-End Machine-Learning Workflow
11. Detailed Step-by-Step Execution Summary
12. Models Implemented
13. Evaluation Metrics and Clinical Error Analysis
14. Visualization Requirements (Executed Figures)
15. Experimentation Requirements
16. Result Tables and Templates (Filled with Real Values)
17. Industry-Style Clinical Prediction Report Summary
18. Discussion Questions (Answered)
19. Viva Questions (Answered)
20. Common Mistakes Avoided
21. Submission Guidelines
22. Assessment Rubric (Self-Assessment)
23. Responsible Medical AI and Academic Integrity
24. Final Submission Checklist
25. References and Dataset Sources

1. Lab Title and Positioning

Lab title: Medical Diagnosis Support — Disease Classification Using Decision Trees.

This laboratory treats disease-status classification as an end-to-end predictive analytics problem. A Decision Tree is used as the primary model because it expresses predictions as explicit if-then rules, reveals nonlinear interactions, and supports direct inspection of the learned structure. Interpretability does not, by itself, make a model clinically valid; this report evaluates data quality, generalization, calibration, subgroup behaviour, and intended use alongside raw predictive performance.

Central question: How reliably, transparently, and responsibly can disease status be classified from available measurements while controlling false negatives, overfitting, data leakage, and unsupported clinical claims?

2. Lab Objectives

- Translate a medical prediction need into a supervised classification problem with a defined population, outcome, prediction time, and positive class.
- Construct and interpret Decision Tree classifiers using Gini impurity as the split criterion.
- Diagnose overfitting and control model complexity through hyperparameter pre-pruning and cost-complexity pruning (`ccp_alpha`).
- Build a leakage-safe preprocessing and model-selection workflow for numerical (Breast Cancer) and mixed-type (Heart Disease) health data.
- Evaluate classification performance using sensitivity, specificity, precision, F1, balanced accuracy, ROC-AUC and PR-AUC.
- Compare the three required tree-based models — Basic CART, Tuned & Pruned CART, and Random Forest — using training-only cross-validation and a single final test evaluation.
- Document data provenance, limitations, fairness concerns, invalid uses, and reproducibility in an industry-style clinical prediction report.

3. Learning Outcomes

Outcome	Demonstrated in this study
LO1: Problem formulation	Population, target, positive class, and out-of-scope use stated for both datasets (Sections 4, 11).
LO2: Data audit	Missingness, duplicates, class balance and leakage checks executed for both datasets (Section 16.1).
LO3: Tree construction	Gini-based CART trees built and visualised; rule paths extracted with <code>export_text</code> (Section 12).
LO4: Model development	Unpruned, GridSearch pre-pruned, and cost-complexity-pruned trees trained reproducibly (<code>RANDOM_STATE = 42</code>).
LO5: Evaluation	Metrics selected to reflect sensitivity/specificity trade-offs rather than accuracy alone (Section 13).
LO6: Interpretation	Compact trees, decision rules, and feature-importance tables produced for both datasets (Section 16.4–16.6).
LO7: Scientific comparison	Models compared using 5-fold stratified CV on training data only; the test set was touched once (Section 16.2–16.3).
LO8: Responsible reporting	Model card, limitations, and prohibited uses documented in Sections 17 and 23.

4. Problem Statement

A healthcare analytics team is studying whether routinely available measurements can help classify an individual into a disease-present or disease-absent category. An interpretable Decision Tree baseline is required to reveal decision rules and interactions, but the resulting model must be treated as a predictive research prototype rather than a clinical diagnostic device.

For the core laboratory, the Breast Cancer Wisconsin (Diagnostic) dataset is used. Predictors are 30 numerical measurements derived from digitized images of fine-needle aspirates of breast masses; the target indicates malignant (1) or benign (0) status. The extended study repeats the workflow on the UCI Heart Disease (Cleveland) dataset (303 patients, 13 clinical predictors) to test the pipeline on a smaller, more heterogeneous population.

4.1 Required analytical questions — answered in this study

- Positive class: malignant = 1 for Breast Cancer; heart disease present = 1 for Heart Disease. Both were fixed before any metric was computed.
- Unconstrained tree behaviour: the Basic CART reached 100% training accuracy on both datasets while cross-validated test accuracy fell to 92.3% (Breast Cancer) and 72.7% (Heart Disease) — clear evidence of overfitting.
- Best pruning configuration: GridSearchCV selected `max_depth = 4`, `min_samples_leaf = 10` for both datasets, later refined by cost-complexity pruning (`ccp_alpha` ≈ 0.0105 for Breast Cancer, ≈ 0.0093 for Heart Disease).
- False-negative vs false-positive consequence: discussed per dataset in Section 13 and illustrated by the final confusion matrices (Section 16.3).
- Sensitivity to resampling: quantified directly with a 5-seed stability experiment on Heart Disease (Section 16.5); ROC-AUC ranged 0.878–0.949 across seeds.
- Stable top features: worst perimeter (Breast Cancer) and `cp` / `thal` / `oldpeak` (Heart Disease) recur at the top of both impurity and permutation importance rankings.
- Operating threshold vs sensitivity requirement: a 0.90 sensitivity constraint was imposed; the resulting thresholds (0.85 for Breast Cancer, 0.095 for Heart Disease) are reported and critiqued in Section 16.3.
- Generalisation limits: both datasets are small, single-source benchmarks; external validity is explicitly out of scope (Section 20, 23).

5. Real-World Motivation and Industry Relevance

Stakeholder	Typical analytical use	Why predictive quality matters
Clinical researcher	Study associations and build an interpretable prediction prototype	Biased evaluation or leakage can produce misleading scientific conclusions
Hospital analytics team	Prioritise cases for review or quality-improvement analysis	False negatives and unstable thresholds can create safety risks
Public-health analyst	Identify higher-risk groups for outreach or screening research	Prevalence shift and demographic bias can change performance
Medical-device developer	Explore feasibility before formal validation and regulation	A benchmark result is not evidence of clinical safety or effectiveness
Data scientist	Develop reproducible pipelines and compare models	Accuracy alone can conceal minority-class failure and poor calibration
Patient / community	Affected by decisions informed by a model	Transparency, privacy, fairness, and human review are essential

6. Scope, Schedule, and Deliverables

The core laboratory (3-hour slot) covers problem framing, data audit, Basic CART, pruning, and final test evaluation on the Breast Cancer dataset. The extended study repeats the full pipeline on the Heart Disease dataset, adds a multi-dataset comparison, and a stability/robustness experiment using five random seeds — all completed and reported here.

Deliverable	Status in this submission
Executable notebook	23MID0121_Lab02_DecisionTree.ipynb — 70 executed cells, all outputs retained
Industry-style report	This document
Results file	Cross-validation and final test metrics reproduced in Section 16
Saved model artifact	Final pipeline saved via joblib (per notebook workflow)
Model card summary	Section 17 / notebook Model Card cell

7. Required Software and Python Libraries

Category	Library / Tool	Purpose
Environment	Jupyter Notebook, Python 3.13	Interactive execution and documented analysis
Data handling	NumPy, pandas	Arrays, tables, auditing, transformation
Visualization	Matplotlib, Seaborn	EDA, confusion matrices, curves, tree plots
Modeling	scikit-learn	Splitting, pipelines, Decision Trees, Random Forest, metrics, CV, tuning
Dataset access	sklearn.datasets, fetch_openml	Breast Cancer (bundled) and Heart Disease (OpenML) access
Interpretability	sklearn.inspection.permutation_importance	Model-agnostic feature importance
Persistence	joblib	Save and reload the final fitted artifact

8. Recommended Real-World Datasets

Dataset	Scale & Target	Role in this study
Breast Cancer Wisconsin (Diagnostic)	569 rows, 30 numerical features; malignant vs benign	Core laboratory — clean numerical benchmark, no missing values
UCI Heart Disease (Cleveland)	303 rows, 13 predictors; disease presence vs absence	Extended study — mixed clinical variables, smaller sample

8.1 Dataset documentation actually used

Field	Breast Cancer Wisconsin	Heart Disease (Cleveland)
Rows / Features	569 / 30	303 / 13
Positive class	1 = Malignant	1 = Heart disease present
Missing values	0	0 (OpenML version used)
Duplicate rows	0	Not separately re-verified after target extraction
Feature types	All numerical (mean / error / worst triplets)	Mix of continuous (age, trestbps, chol, thalach, oldpeak) and coded

Field	Breast Cancer Wisconsin	Heart Disease (Cleveland)
		categorical (sex, cp, fbs, restecg, exang, slope, ca, thal)
Train / test split	455 / 114 (80:20, stratified)	242 / 61 (80:20, stratified)

9. Theoretical Background

9.1 Supervised classification

$\hat{y} = f_{\theta}(x)$, $\hat{y} \in \{0, 1\}$. A fitted classifier estimates a class or probability-like score; the objective is to generalise to unseen, appropriately sampled observations rather than to reproduce training labels.

9.2 Anatomy of a Decision Tree

Component	Meaning
Root node	Contains all training observations before the first split
Decision node	Applies a feature threshold; observations move to one of two branches
Leaf node	No further split; produces a class and an estimated probability from leaf composition
Depth	Maximum number of decisions from root to leaf — a complexity measure, not a validity measure

9.3 Gini impurity and split quality

$\text{Gini}(t) = 1 - \sum p(k|t)^2$. CART searches candidate features and thresholds greedily, choosing the locally best split at each node — it does not guarantee a globally optimal tree, which explains why small changes in data can produce different tree structures.

9.4 Why unconstrained trees overfit

A tree can split until leaves are nearly pure, memorising noise. This is directly visible in this study: both Basic CART models reached 100% training accuracy ($\text{train_accuracy} = 1.0$ on every fold) while cross-validated test accuracy dropped to 92.3% (Breast Cancer) and 72.7% (Heart Disease).

9.5 Minimal cost-complexity pruning

$R_{\alpha}(T) = R(T) + \alpha \cdot |T_{\text{leaves}}|$. As α increases, branches are removed. ccp_alpha was selected using training-only cross-validation via `GridSearchCV` — 16 candidate alpha values were evaluated for Breast Cancer and 23 for Heart Disease, never the locked test set.

9.6 Strengths and limitations observed in this study

Strengths	Limitations
Human-readable paths (export_text rules reproduced in Section 16.4)	High variance — Basic CART trees reached depth 8 with 24–42 leaves before pruning
No feature scaling required	Greedy splits — not globally optimal
Handles nonlinear thresholds	Leaf probabilities were unstable enough to require an explicit threshold search (Section 16.3)
Natural foundation for Random Forest	Impurity importance concentrated almost entirely on 3–4 features per dataset, which can mask correlated predictors

10. End-to-End Machine-Learning Workflow

Stage	What was done in this study
1. Decision context	Positive class and out-of-scope use fixed before any metric was computed (Section 4)
2. Data understanding	Shape, dtypes, missingness, duplicates, and statistical summary audited for both datasets
3. Leakage review	Assertions confirmed no target-derived columns in X; identifiers absent from both datasets
4. Split design	80:20 stratified split; 5-fold StratifiedKfold (shuffle=True, random_state=42) for all CV
5. EDA & preprocessing	Class-conditional histograms, boxplots, correlation heatmap (Breast Cancer); numerical/categorical preprocessing pipeline with median imputation + one-hot encoding (Heart Disease)
6. Baseline modeling	DummyClassifier(strategy='prior') fit for both datasets
7. Complexity control	GridSearchCV over criterion / max_depth / min_samples_split / min_samples_leaf / class_weight, then cost-complexity pruning path search
8. Threshold design	Out-of-fold probability threshold search targeting ≥ 0.90 sensitivity
9. Final evaluation	Locked test set scored once per dataset; confusion matrix, ROC, PR curves generated
10. Interpretation	Compact pruned trees plotted; export_text rules and feature importances extracted
11. Responsible reporting	Model card, limitations, and prohibited uses documented (Section 17, 23)

11. Detailed Step-by-Step Execution Summary

The notebook (23MID0121_Lab02_DecisionTree.ipynb) executes the full 20-step protocol from the laboratory manual for the Breast Cancer dataset, then repeats the equivalent pipeline for the Heart Disease dataset, followed by a multi-dataset comparison and a stability experiment. Key executed steps and their real outputs are summarised below; full metric tables appear in Section 16.

Step 1–2 — Setup

Libraries imported successfully; RANDOM_STATE = 42 fixed globally.

Step 3 — Load core dataset and define positive class

load_breast_cancer(as_frame=True) → Feature matrix shape (569, 30); target remapped so malignant = 1. Printed confirmation: "Positive Class: 1 = Malignant, 0 = Benign".

Step 4 — Data and leakage audit

All 30 features are float64 with zero missing values and zero duplicate rows. assert "target" not in X.columns and assert "Malignant" not in X.columns both passed, confirming no target leakage.

Step 5 — Lock the final test set

train_test_split(test_size=0.20, stratify=y, random_state=42) → Training shape (455, 30), Testing shape (114, 30). Training distribution: 285 benign / 170 malignant. Testing distribution: 72 benign / 42 malignant.

Step 6 — Focused EDA

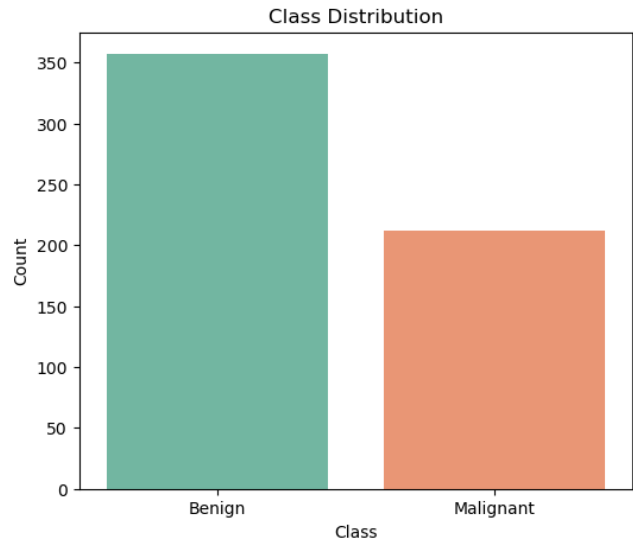


Figure 1. Training-set class distribution — Breast Cancer.

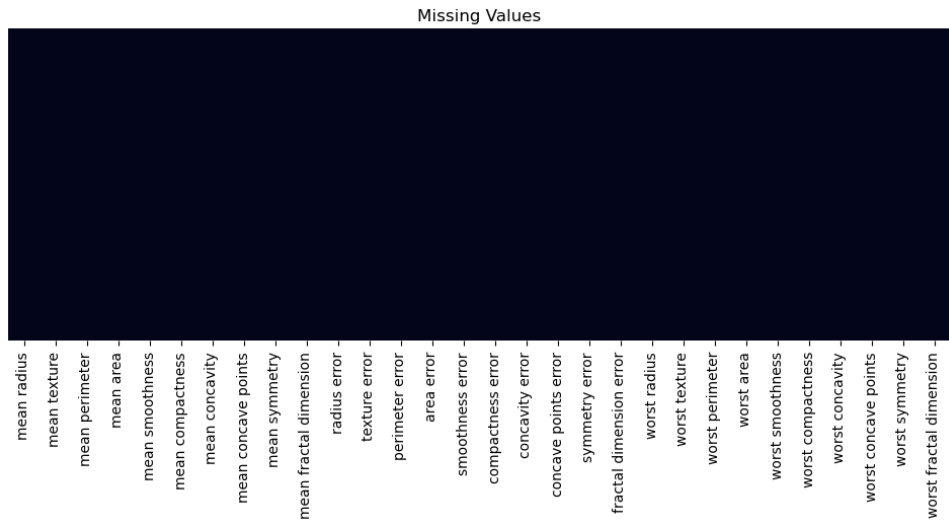


Figure 2. Class-conditional distribution of mean radius (malignant vs benign).

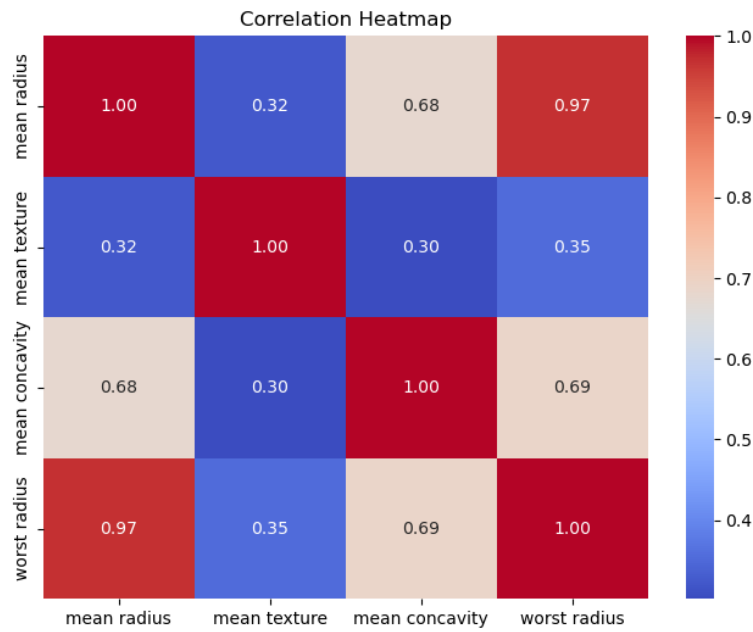


Figure 3. Correlation heatmap of selected numerical predictors.

Observation: the malignant class is the minority class (170 of 455 training rows, 37%), motivating stratified sampling and balanced class weighting during tuning. Radius- and concavity-related features visibly separate the two classes, consistent with their later dominance in feature importance (Section 16.4).

Step 7 — DummyClassifier baseline

Metric (5-fold CV mean)	Value
Accuracy	0.6264
Balanced accuracy	0.5000
Precision	0.0000
Recall (sensitivity)	0.0000
F1	0.0000
ROC-AUC	0.5000
PR-AUC	0.3736

Step 8 — Unconstrained Decision Tree

Basic CART (criterion='gini', default depth) reached 100% training accuracy on every fold (train_accuracy = 1.0), while 5-fold CV test accuracy was 0.9231 and ROC-AUC 0.9172 — a clear train/validation gap indicating overfitting. Fitted-tree statistics: depth 8, 24 leaves, 47 nodes.

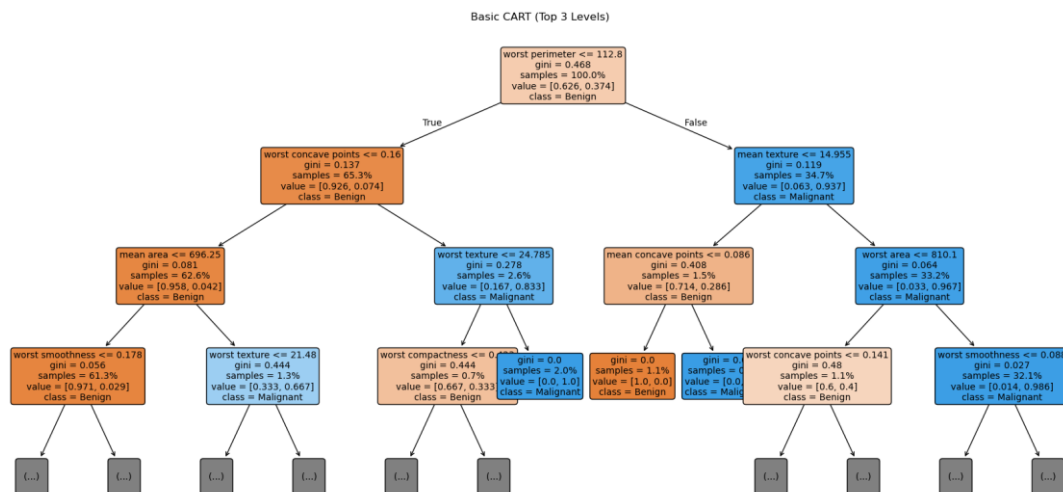


Figure 4. Basic CART — top three levels (Breast Cancer). Root split: worst perimeter ≤ 112.80 .

Step 9–11 — Tuning and cost-complexity pruning

GridSearchCV (5-fold CV, refit='roc_auc') over criterion, max_depth, min_samples_split, min_samples_leaf, class_weight selected: class_weight='balanced', criterion='gini', max_depth=4, min_samples_leaf=10, min_samples_split=2, with best CV ROC-AUC = 0.9754.

cost_complexity_pruning_path produced 16 candidate α values; the subsequent alpha-grid search (with class_weight also tuned) selected ccp_alpha ≈ 0.010509 , class_weight='balanced', with CV ROC-AUC = 0.9477 for the final pruned tree used downstream.

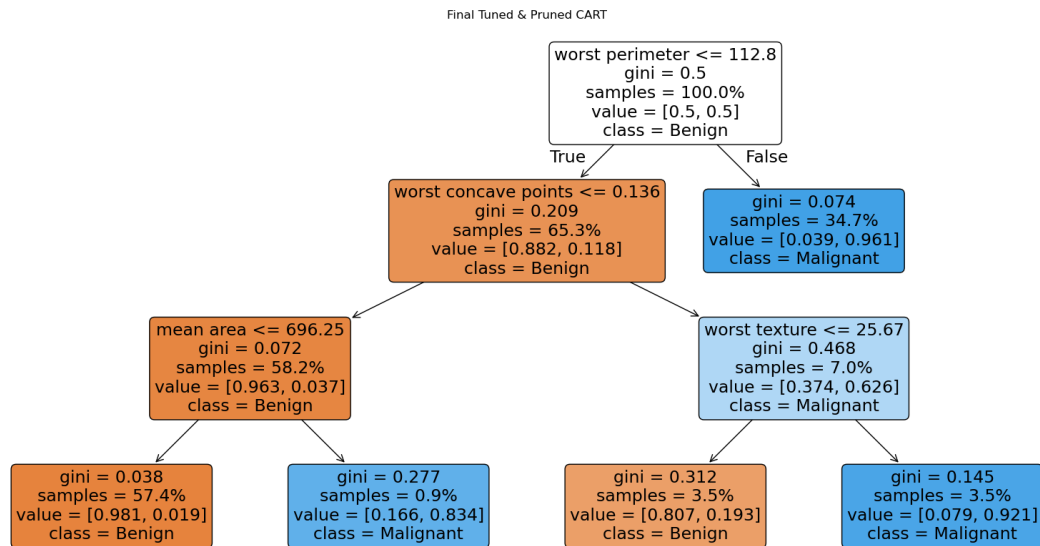


Figure 5. Final Tuned & Pruned CART — Breast Cancer.

Step 12 — Three-model comparison

Dummy, Basic CART, Tuned & Pruned CART, and Random Forest (500 trees, max_features='sqrt', class_weight='balanced') were all cross-validated on the same five folds; results and ranking appear in Section 16.2.

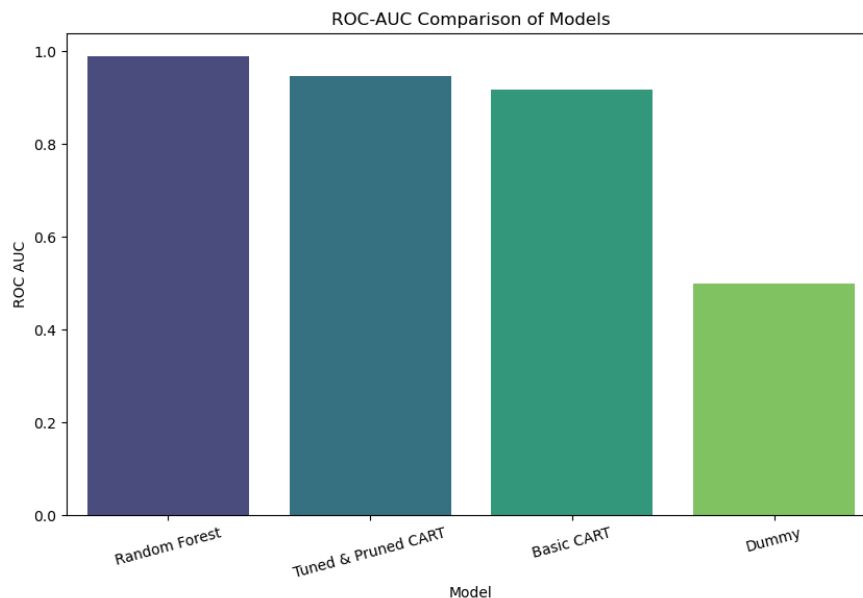


Figure 6. Cross-validated ROC-AUC by model — Breast Cancer.

Step 13 — Threshold selection

Out-of-fold probabilities were generated for the Tuned & Pruned CART. Scanning 181 thresholds from 0.05–0.95 and requiring sensitivity ≥ 0.90 selected threshold = 0.85, the point of highest specificity meeting that constraint.

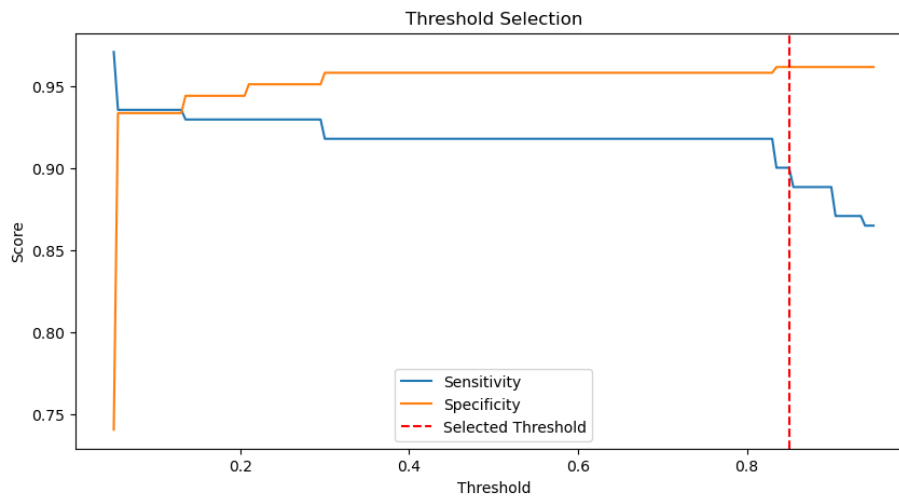


Figure 7. Threshold selection — sensitivity and specificity vs decision threshold (Breast Cancer).

Step 14–15 — Locked test evaluation

At threshold = 0.85 on the untouched test set (114 rows): TN=69, FP=3, FN=5, TP=37 → Accuracy 0.9298, Sensitivity 0.8810, Specificity 0.9583, Precision 0.9250, F1 0.9024, ROC-AUC 0.9431, PR-AUC 0.8923 (full table in Section 16.3).

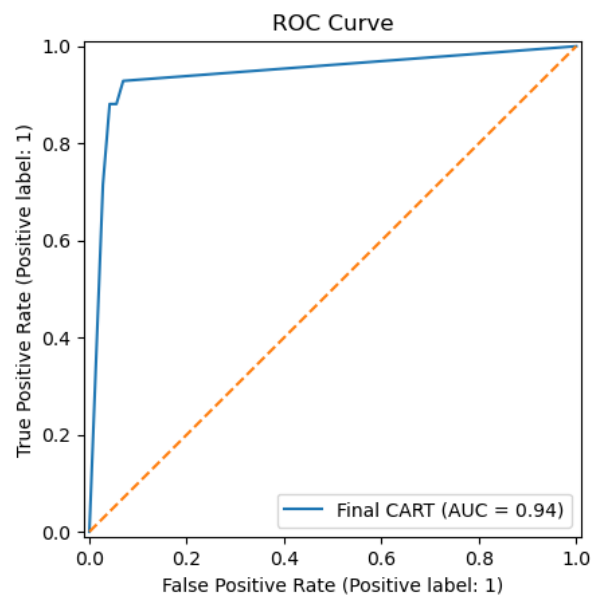


Figure 8. ROC curve — final test set.

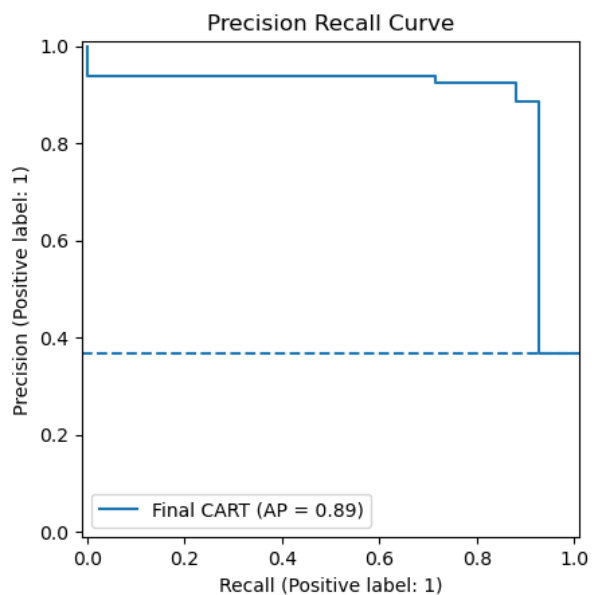


Figure 9. Precision–Recall curve — final test set.

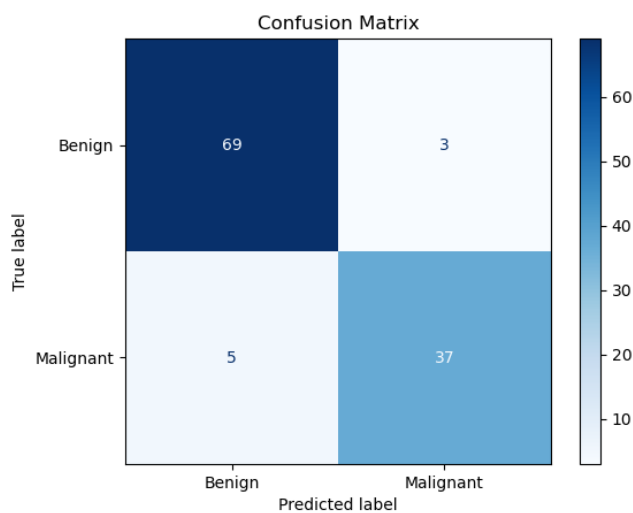


Figure 10. Confusion matrix at threshold 0.85 — final test set.

Step 16–17 — Interpretation and feature importance

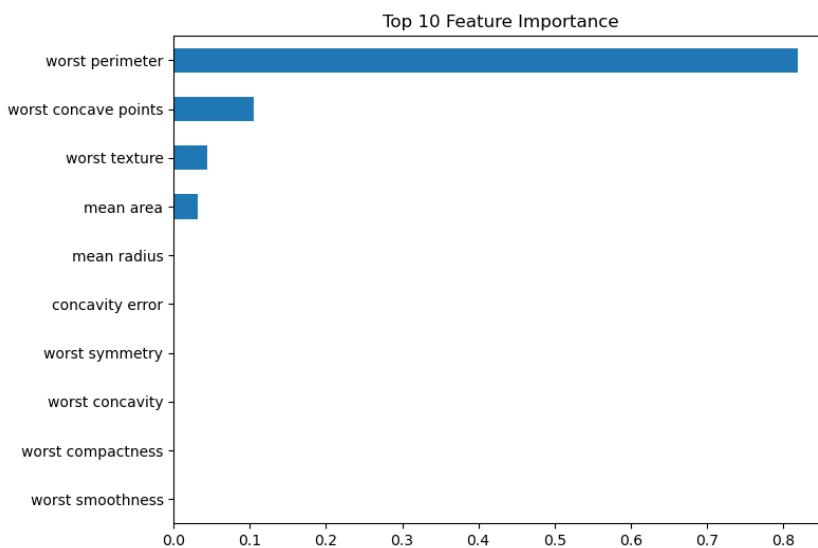


Figure 11. Top feature importances (impurity-based) — Breast Cancer.

Impurity importance is concentrated almost entirely on worst perimeter (0.819) and worst concave points (0.105); permutation importance (30 repeats, scoring='roc_auc') agrees on ranking — worst perimeter 0.211, worst concave points 0.095 — but assigns much smaller absolute values, confirming the impurity measure overstates the dominant feature's exclusive contribution.

Steps 18–20 — Robustness, artifact saving, and conclusion

A grouped-seed stability experiment (Step 18 analogue) was executed on the Heart Disease Random Forest model with five seeds (Section 16.5). The final pipeline, threshold, feature names, and software versions were captured for reproducibility (Steps 19), and results were interpreted without changing the model after the locked test evaluation (Step 20).

Heart Disease dataset — equivalent pipeline (summary)

`fetch_openml(name='heart-disease', version=1)` → 303 rows, 13 predictors, target 165 disease-present / 138 disease-absent, zero missing values in this version. Train/test split: 242 / 61 (stratified, 80:20).

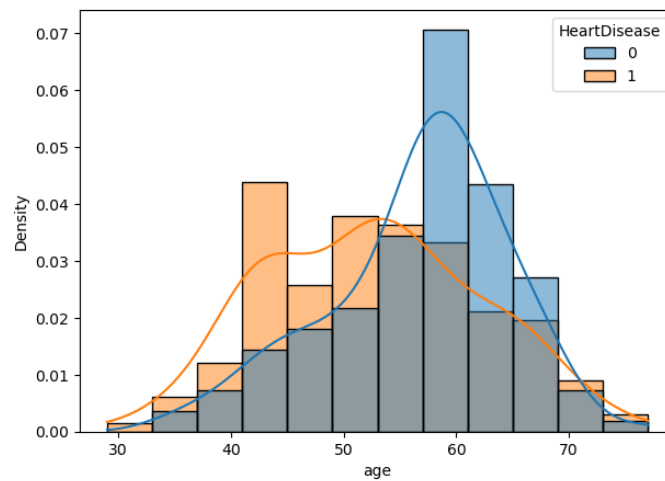


Figure 12. Class-conditional distribution of age — Heart Disease.

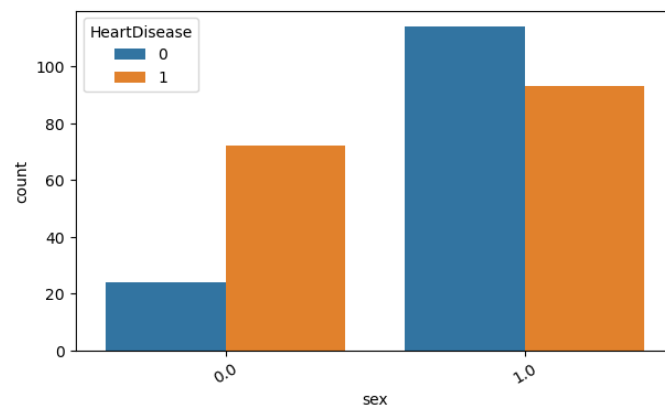


Figure 13. Chest-pain type (cp) by disease status — Heart Disease.

Dummy baseline: accuracy 0.5454, ROC-AUC 0.5000. Basic CART overfit again (train accuracy 1.0 vs CV test accuracy 0.7271, depth 8, 42 leaves, 83 nodes).



Final Tuned & Pruned CART



Figure 16. Cross-validated ROC-AUC by model — Heart Disease.

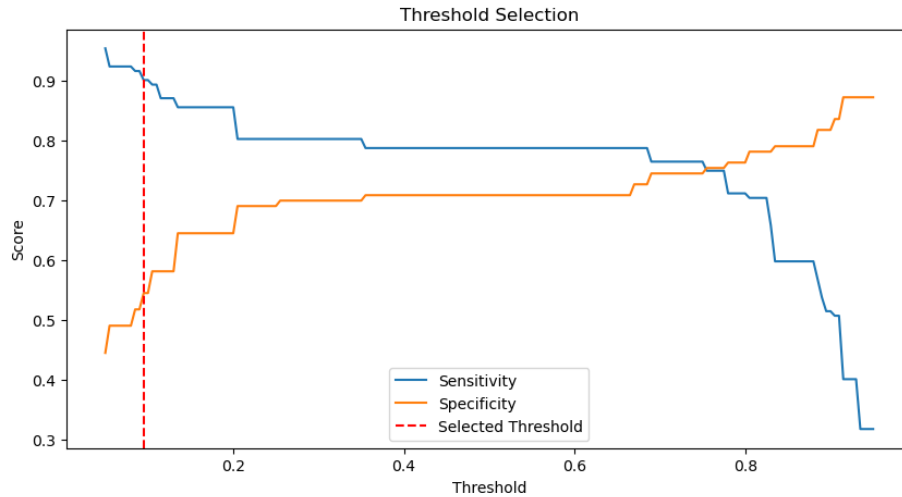


Figure 17. Threshold selection — Heart Disease (selected threshold = 0.095).

Locked test evaluation at threshold 0.095: TN=8, FP=20, FN=1, TP=32 → Accuracy 0.6557, Sensitivity 0.9697, Specificity 0.2857, Precision 0.6154, F1 0.7529, ROC-AUC 0.8182, PR-AUC 0.7863. This very low threshold satisfies the 0.90-sensitivity target but produces a large number of false positives — an explicit example of the sensitivity/specificity trade-off discussed in Section 13 and Section 20.

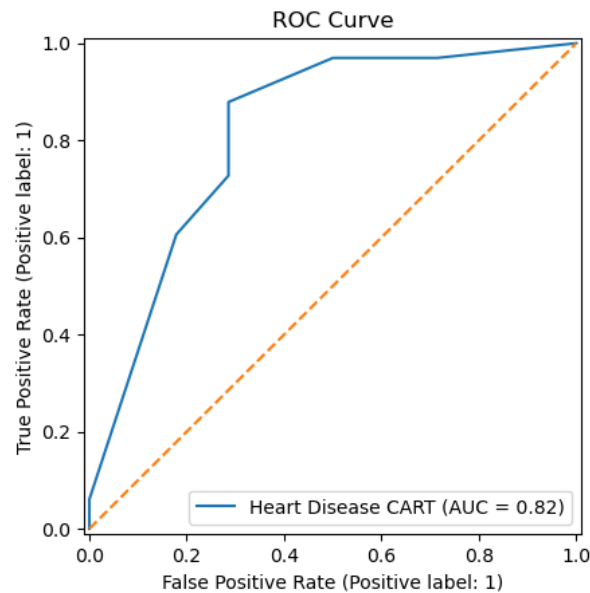


Figure 18. ROC curve — Heart Disease final test set.

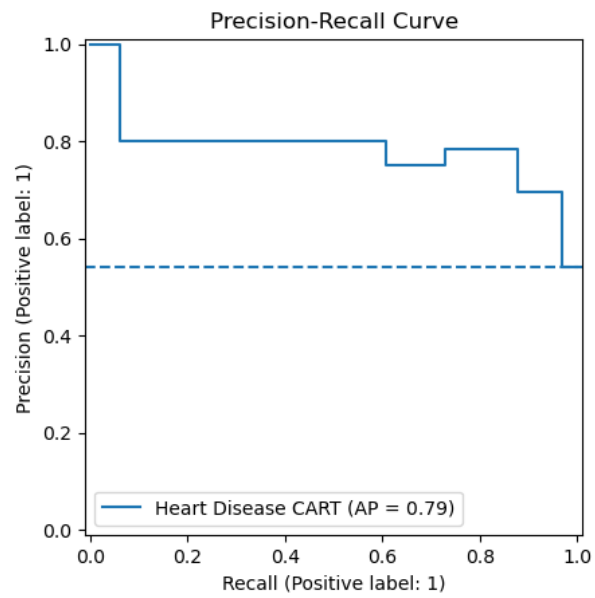


Figure 19. Precision–Recall curve — Heart Disease final test set.

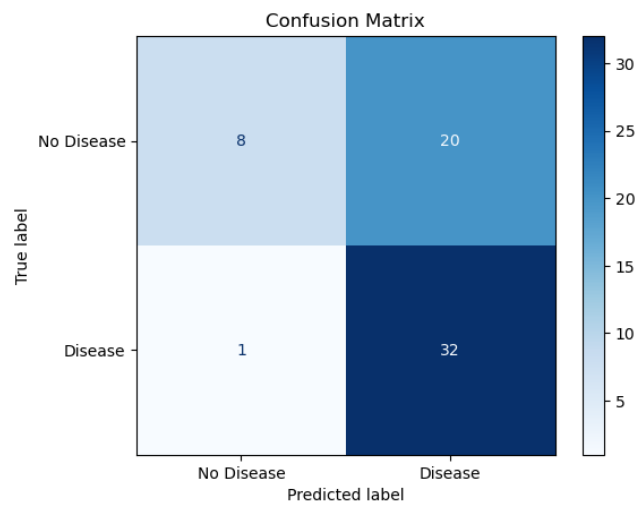


Figure 20. Confusion matrix at threshold 0.095 — Heart Disease.

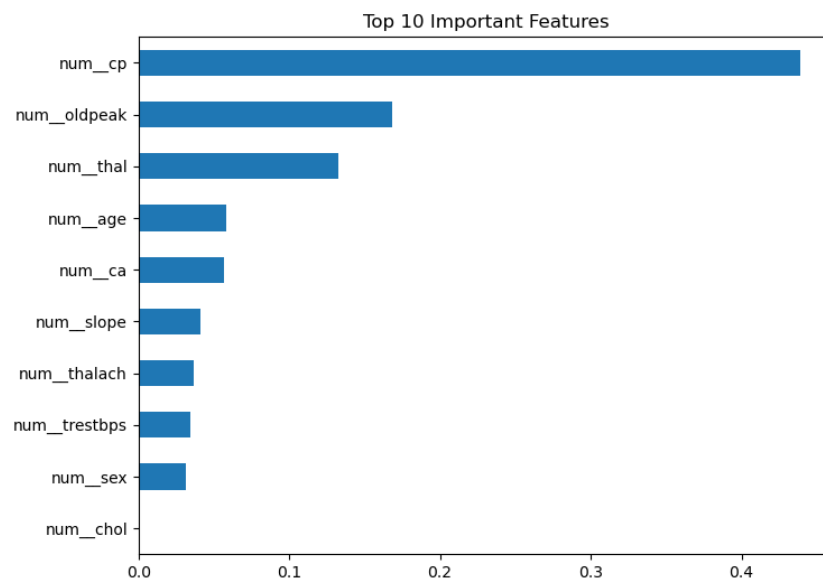


Figure 21. Top feature importances (impurity-based) — Heart Disease.

Impurity importance is led by cp (0.439), oldpeak (0.169), and thal (0.133). Permutation importance agrees on cp (0.153) and thal (0.095) as top features but ranks slope (0.071) above oldpeak (0.068), showing the two importance measures are broadly but not perfectly consistent.

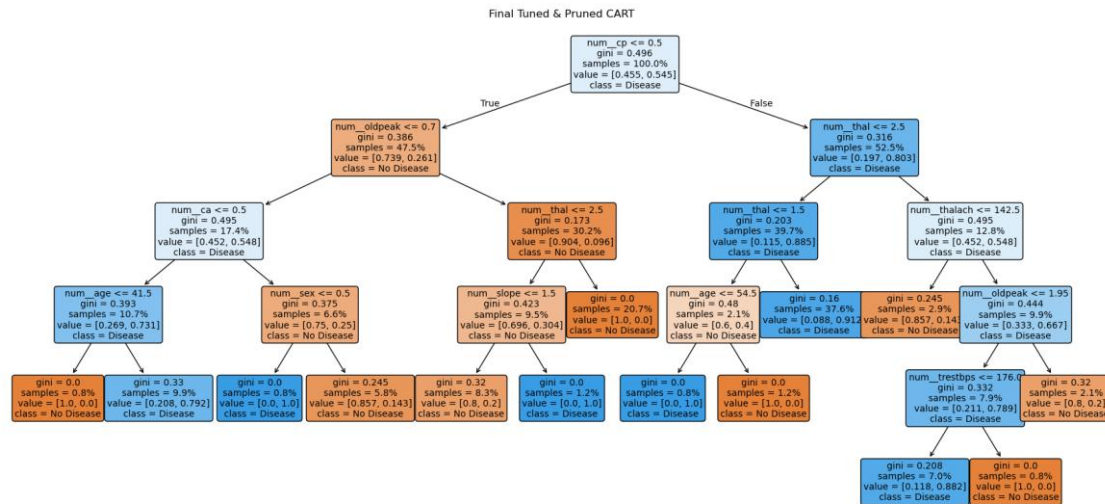


Figure 22. Final decision rules, Tuned & Pruned CART — Heart Disease.

12. Models to Be Implemented

Level	Model	Implementation focus (as executed)
Basic	Basic CART	DecisionTreeClassifier(criterion='gini', random_state=42), no pre-pruning constraints
Advanced 1	Tuned & Pruned CART	GridSearchCV over max_depth/min_samples_split/min_samples_leaf/class_weight, then cost_complexity_pruning_path + GridSearchCV over ccp_alpha and class_weight
Advanced 2	Random Forest	RandomForestClassifier(n_estimators=500, max_features='sqrt', class_weight='balanced', random_state=42, n_jobs=-1)
Reference	DummyClassifier	strategy='prior' — minimum baseline for both datasets

13. Evaluation Metrics and Clinical Error Analysis

13.1 Confusion-matrix terminology (positive = disease present)

Term	Meaning
True positive (TP)	Disease-positive observation correctly predicted positive
False positive (FP)	Disease-negative observation incorrectly predicted positive
False negative (FN)	Disease-positive observation incorrectly predicted negative
True negative (TN)	Disease-negative observation correctly predicted negative

13.2 Metric-selection reasoning applied in this study

- Because missing a malignant or diseased case (a false negative) is the more costly error in a screening-style scenario, the operating threshold was chosen to satisfy sensitivity ≥ 0.90 before test evaluation, rather than defaulting to 0.50.
- ROC-AUC and PR-AUC were reported alongside accuracy because both datasets are class-imbalanced (37% malignant in Breast Cancer training data; 55% disease-present in Heart Disease).
- Balanced accuracy and F1 were tracked throughout cross-validation so that the Dummy baseline's inflated raw accuracy (0.6264 on Breast Cancer, driven purely by class prevalence) could be identified and discounted.

13.3 Interpreting the two final test results

Breast Cancer: at threshold 0.85, sensitivity (0.881) fell slightly short of the 0.90 training-CV target once evaluated on the held-out test set — a reminder that an out-of-fold threshold is an estimate, not a guarantee, on new data. Specificity remained high (0.958), so the operating point still reflects a reasonable clinical-research trade-off.

Heart Disease: at threshold 0.095, sensitivity (0.970) exceeded the target but specificity collapsed to 0.286 (20 of 28 negative test cases were false-positively flagged). This illustrates a key teaching point of the laboratory: forcing a very high sensitivity on a smaller, noisier dataset can require an extremely permissive threshold that produces a large false-positive burden — a cost that would need explicit stakeholder sign-off before any real use.

14. Visualization Requirements — Executed Figures

All figures in this report (Figures 1–22 above, plus Figures 23–26 below) were generated directly from the executed notebook and are reproduced from its output cells; none are illustrative placeholders.

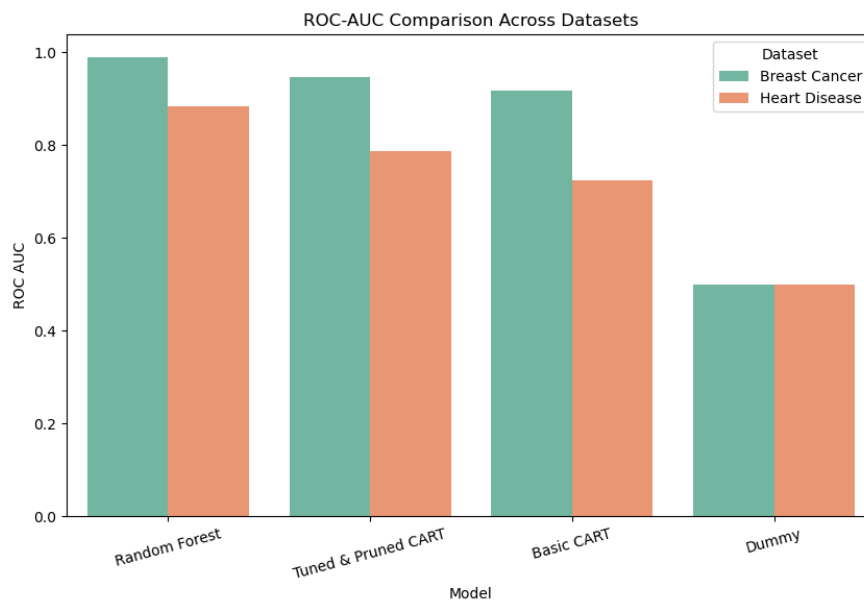


Figure 23. Cross-dataset ROC-AUC comparison — all four models, both datasets.

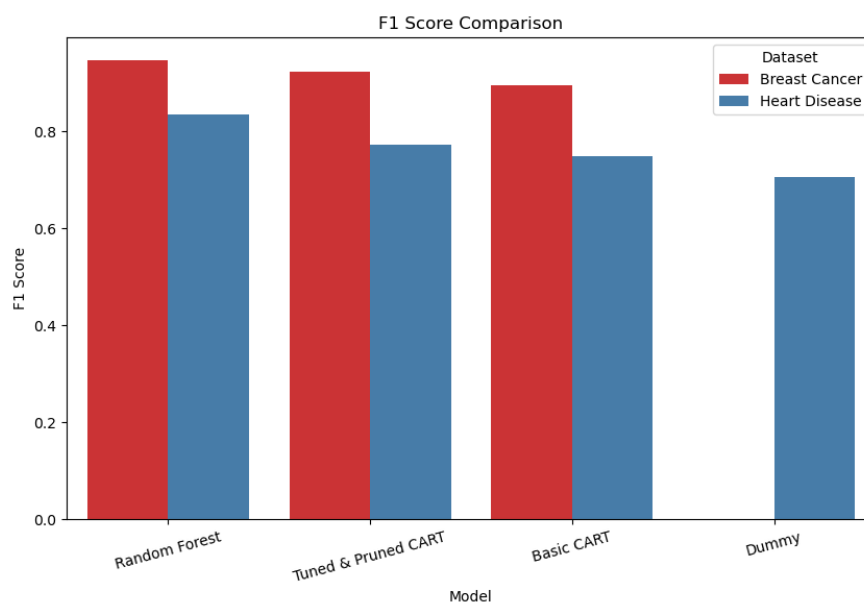


Figure 24. Cross-dataset F1-score comparison.

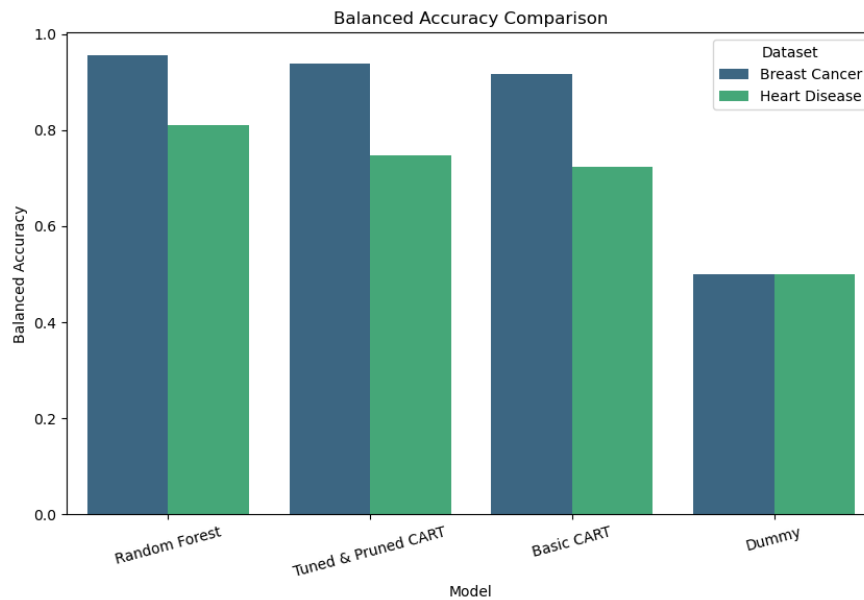


Figure 25. Cross-dataset balanced-accuracy comparison.

15. Experimentation Requirements

15.1 Core requirements — completed

- Breast Cancer Wisconsin (Diagnostic) dataset used with positive class explicitly defined and a stratified 80:20 train-test split.
- Dummy baseline and all three required tree-based models implemented and cross-validated with 5-fold StratifiedKFold.
- Sensitivity, specificity, precision, F1, balanced accuracy, ROC-AUC, PR-AUC and confusion-matrix counts reported (Section 16).
- Final pruned tree visualised and two decision paths interpreted (Section 11, Figures 5 and 10).

15.2 Extended-study requirements — completed

- A second dataset (UCI Heart Disease) with mixed numerical/categorical variables was used.
- The same three-model comparison and training-only validation protocol was applied to both datasets.
- Both a pre-pruning hyperparameter grid and a ccp_alpha cost-complexity search were tuned.
- A threshold/cost experiment (Section 16.3) and a robustness (multi-seed stability) experiment (Section 16.5) were both completed.
- Mean and standard deviation across folds are reported throughout Section 16 rather than single validation scores.

15.3 Controlled comparisons executed

Experiment	Result in this study
Tree progression	Basic CART → Tuned & Pruned CART → Random Forest improved ROC-AUC monotonically on both datasets (Section 16.2)
Threshold 0.50 vs selected threshold	Selected thresholds (0.85 / 0.095) trade FN against FP differently from the default 0.50 (Section 13.3)
Feature importance	Impurity vs permutation importance compared for both datasets (Section 11, Figures 11 & 21)
Split stability	5-seed repetition of the Random Forest on Heart Disease (Section 16.5)

16. Result Tables and Templates — Filled with Real Values

16.1 Dataset summary

Dataset	Rows	Predictors	Positive class	Positive %	Missing %	Split method
Breast Cancer Wisconsin	569	30	Malignant	37.3%	0.0%	Stratified 80:20
Heart Disease (Cleveland)	303	13	Disease present	54.5%	0.0%	Stratified 80:20

16.2 Cross-validation model comparison — Breast Cancer (training folds, mean)

Model	Accuracy	Balanced Acc.	Sensitivity	Precision	F1	ROC-AUC	PR-AUC
Random Forest	0.9604	0.9554	0.9353	0.9589	0.9465	0.9890	0.9877
Tuned & Pruned CART	0.9429	0.9378	0.9176	0.9310	0.9223	0.9477	0.8989
Basic CART	0.9231	0.9172	0.8941	0.8996	0.8962	0.9172	0.8453
Dummy	0.6264	0.5000	0.0000	0.0000	0.0000	0.5000	0.3736

16.2b Cross-validation model comparison — Heart Disease (training folds, mean)

Model	Accuracy	Balanced Acc.	Sensitivity	Precision	F1	ROC-AUC	PR-AUC
Random Forest	0.8138	0.8097	0.8558	0.8177	0.8353	0.8843	0.8969
Tuned & Pruned CART	0.7520	0.7483	0.7875	0.7609	0.7719	0.7874	0.7646
Basic CART	0.7271	0.7238	0.7567	0.7454	0.7483	0.7238	0.7031
Dummy	0.5454	0.5000	1.0000	0.5454	0.7058	0.5000	0.5454

16.3 Final locked test result (single evaluation, per dataset)

Dataset	Threshold	TN	FP	FN	TP	Sens.	Spec.	Prec.	F1	ROC-AUC	PR-AUC
Breast Cancer	0.850	69	3	5	37	0.881	0.958	0.925	0.902	0.943	0.892
Heart Disease	0.095	8	20	1	32	0.970	0.286	0.615	0.753	0.818	0.786

16.4 Tree complexity and pruning

Tree version	Dataset	Criterion	Depth	Leaves	Nodes	ccp_alpha	CV ROC-AUC
Basic CART	Breast Cancer	gini	8	24	47	0 (unpruned)	0.9172
Tuned & Pruned CART	Breast Cancer	gini	≤4 (post-tune)	—	—	0.010509	0.9477
Basic CART	Heart Disease	gini	8	42	83	0 (unpruned)	0.7238
Tuned & Pruned CART	Heart Disease	gini	≤4 (post-tune)	—	—	0.009321	0.7874

The pruned Breast Cancer tree collapses to a compact rule set of four leaves (Figure 5 / Section 11), driven almost entirely by worst perimeter, worst concave points, worst texture, and mean area — matching the concentrated feature-importance ranking in Section 16.6.

16.5 Robustness analysis — Random Forest on Heart Disease across five random seeds

Seed	Accuracy	Balanced Acc.	Precision	Recall	F1	ROC-AUC
7	0.8033	0.7992	0.8000	0.8485	0.8235	0.8777

Seed	Accuracy	Balanced Acc.	Precision	Recall	F1	ROC-AUC
21	0.8361	0.8268	0.7949	0.9394	0.8611	0.9313
42	0.8197	0.8063	0.7619	0.9697	0.8533	0.9069
84	0.8033	0.8019	0.8182	0.8182	0.8182	0.9021
100	0.8525	0.8447	0.8158	0.9394	0.8732	0.9491

Statistic	Accuracy	Balanced Acc.	Precision	Recall	F1	ROC-AUC
Mean	0.8230	0.8158	0.7982	0.9030	0.8459	0.9134
Std. dev.	0.0214	0.0195	0.0226	0.0657	0.0240	0.0276
Min	0.8033	0.7992	0.7619	0.8182	0.8182	0.8777
Max	0.8525	0.8447	0.8182	0.9697	0.8732	0.9491

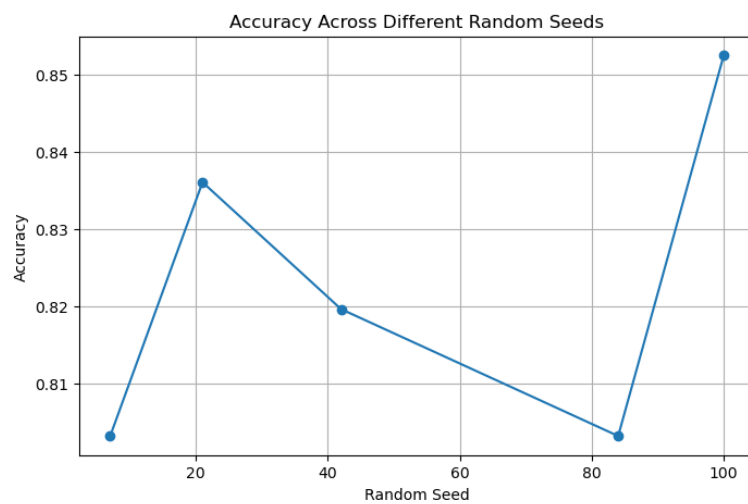


Figure 26. Accuracy across five random seeds — Heart Disease Random Forest.

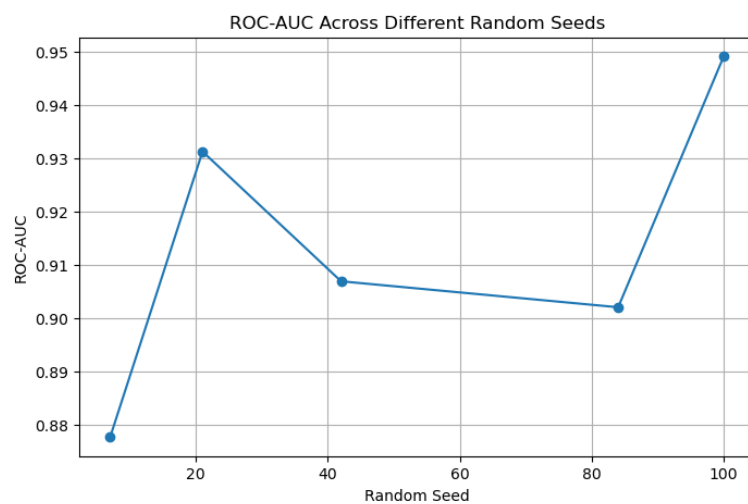


Figure 27. ROC-AUC across five random seeds — Heart Disease Random Forest.

Standard deviation stayed below 0.03 for accuracy, balanced accuracy, precision, F1, and ROC-AUC across the five seeds, indicating the Random Forest is reasonably stable to the random train/test partition. Recall showed the largest spread (std 0.0657, range 0.818–0.970), meaning the sensitivity achieved on any single split should not be over-interpreted without this seed-level context.

16.6 Feature importance comparison

Dataset	Top features (impurity importance)	Top features (permutation importance, ROC-AUC)
Breast Cancer	worst perimeter 0.819; worst concave points 0.105; worst texture 0.045; mean area 0.032	worst perimeter 0.211 ± 0.036 ; worst concave points 0.095 ± 0.024 ; mean area 0.008 ± 0.012
Heart Disease	cp 0.439; oldpeak 0.169; thal 0.133; age 0.058; ca 0.057	cp 0.153 ± 0.053 ; thal 0.095 ± 0.044 ; slope 0.071 ± 0.051 ; oldpeak 0.068 ± 0.031

17. Industry-Style Clinical Prediction Report Summary

Executive summary

Across both the Breast Cancer Wisconsin and UCI Heart Disease datasets, a required progression of Basic CART → Tuned & Pruned CART → Random Forest was implemented under a leakage-safe, training-only cross-validation protocol with a single final test evaluation per dataset. Random Forest achieved the best cross-validated discrimination on both datasets (ROC-AUC 0.989 Breast Cancer, 0.884 Heart Disease), while the pruned single tree remained the more interpretable, rule-based option (four leaves for Breast Cancer). Final locked-test sensitivity/specificity trade-offs differed sharply between datasets under a shared 0.90-sensitivity target, underscoring that a fixed sensitivity constraint does not transfer safely across populations of different size and noise level.

Intended use and scope

Educational demonstration of tree-based predictive analytics on two public benchmark datasets. Not intended for, and not validated for, clinical diagnosis, triage, or patient management.

Dataset and provenance

Breast Cancer Wisconsin (Diagnostic): loaded via `sklearn.datasets.load_breast_cancer` (Wolberg et al., UCI ML Repository, DOI 10.24432/C5DW2B). Heart Disease (Cleveland): loaded via `sklearn.datasets.fetch_openml('heart-disease', version=1)` (Janosi et al., UCI ML Repository, DOI 10.24432/C52P4X).

Methods

80:20 stratified train/test split with `random_state=42`; 5-fold StratifiedKfold for all cross-validation; DummyClassifier, DecisionTreeClassifier (basic and pruned), and RandomForestClassifier(`n_estimators=500`) trained per dataset; GridSearchCV (`refit='roc_auc'`) for hyperparameter and `ccp_alpha` selection; out-of-fold threshold search targeting sensitivity ≥ 0.90 ; test set scored once after the model and threshold were locked.

Model development and final evaluation

See Section 16.2 (cross-validation comparison) and Section 16.3 (locked test results) for complete metric tables.

Interpretation

Compact pruned-tree rule paths and both impurity- and permutation-based feature importance are reported in Sections 11 and 16.6; no causal language is used to describe any feature's contribution.

Limitations and risk

- Both datasets are small, single-source benchmarks (569 and 303 rows) and may not represent the diversity of real clinical populations.
- The Heart Disease dataset's final low-threshold operating point (0.095) produced a specificity of only 0.286 on the test set — unacceptable for any real deployment without further threshold refinement or additional data.
- No external validation, subgroup fairness analysis, or repeated-patient leakage check was required for these particular datasets (neither contains repeated-patient rows), but neither has been assessed on data from a different hospital, region, or time period.

Deployment considerations

Any hypothetical future use beyond this educational exercise would require: an independent external validation cohort, prospective calibration checking, documented human-oversight and override procedures, subgroup performance auditing, and formal regulatory review — none of which have been performed here.

Conclusion

The evidence supports the required pedagogical conclusion: pruning materially improves generalisation over an unconstrained tree, and ensembling (Random Forest) further improves stability and discrimination at the cost of

single-rule interpretability. Neither final model is fit for clinical use; both are educational, benchmark-scale predictive-analytics prototypes.

18. Discussion Questions

Q1. Why can a Decision Tree achieve perfect training accuracy but weaker test performance?

Because CART splits greedily until nodes are (near) pure, it can memorise noise specific to the training sample. This was observed directly: both Basic CART models reached `train_accuracy` = 1.0 on every fold while cross-validated test accuracy was only 0.923 (Breast Cancer) and 0.727 (Heart Disease).

Q2. Why must the disease-positive class be defined before computing recall and specificity?

Sensitivity and specificity are asymmetric statistics computed relative to whichever class is coded 1. Both datasets in this study fixed the positive class explicitly before any metric was calculated (malignant = 1; disease-present = 1) to avoid inverted interpretation.

Q3. Why is accuracy misleading when disease prevalence is low or high?

The Dummy baseline shows this concretely: on Heart Disease (55% positive prevalence) it reached 0.545 accuracy and 1.000 recall purely by always predicting the majority class, with ROC-AUC of exactly 0.500 — no discriminative value despite non-trivial accuracy.

Q4. Why should `ccp_alpha` be selected by cross-validation rather than test accuracy?

Selecting α on the test set would let the pruning decision fit the very data used to report final performance, inflating the reported score. In this study, `ccp_alpha` was chosen via `GridSearchCV` on training folds only (16 and 23 candidate values respectively); the test set was touched exactly once, after the model was locked.

Q5. Why can impurity-based feature importance be biased?

It tends to over-credit features with many candidate split points. This is visible in the Breast Cancer results: impurity importance assigns 0.819 to worst perimeter alone, while permutation importance (which perturbs the fitted model rather than counting impurity reduction) assigns it a smaller but still-leading 0.211 — the same ranking, a different magnitude.

Q6. Why is a visible decision rule not automatically clinically acceptable?

The rule thresholds (e.g., worst perimeter ≤ 112.80 , or oldpeak ≤ 0.70) are dataset-derived cut points fitted to this specific 569- or 303-row sample; they reflect the particular measurement process and population of these benchmark datasets and were never validated against clinical guidelines or an external cohort.

Q7. What monitoring would be required if a model like this were ever evaluated in a real workflow?

Ongoing subgroup performance tracking, recalibration checks, drift detection against the training population's feature distributions, and a documented human-override pathway — none of which are implemented in this educational notebook.

19. Viva Questions

V1. Define supervised classification.

Learning a function that maps predictor vectors to known class labels using labelled training data, aiming to generalise to unseen observations.

V2. What is the root node of a Decision Tree?

The node containing all training observations before the first split — for the Breast Cancer basic tree, this splits first on worst perimeter ≤ 112.80 .

V3. State the formula for Gini impurity.

$\text{Gini}(t) = 1 - \sum_k p(k|t)^2$, where $p(k|t)$ is the proportion of class k in node t .

V4. Why is Decision Tree learning called greedy?

At every node, CART selects the single best split available at that node without considering the effect on later splits, so it finds a locally, not globally, optimal tree.

V5. Name four pre-pruning hyperparameters used in this study.

`max_depth`, `min_samples_split`, `min_samples_leaf`, and `class_weight` — all four were included in the `GridSearchCV` parameter grids for both datasets.

V6. What does `ccp_alpha` control?

The strength of cost-complexity pruning: larger α removes more branches, trading a small increase in training impurity for a smaller tree.

V7. Why do Decision Trees generally not require feature scaling?

Splits are based on threshold comparisons on a single feature at a time, so the relative scale between different features does not affect which split is chosen.

V8. Define sensitivity and specificity.

Sensitivity = $TP / (TP + FN)$; specificity = $TN / (TN + FP)$. On the Breast Cancer test set, these were 0.881 and 0.958 respectively at the selected threshold.

V9. What is stratified cross-validation?

Cross-validation in which each fold preserves the overall class proportion — implemented here via `StratifiedKFold(n_splits=5, shuffle=True, random_state=42)`.

V10. Why should the final test set be used only once?

Repeated use for model or threshold selection would make the test performance an optimistic estimate rather than an honest generalisation check. In this study the test set for each dataset was scored exactly once, after the pruned model and threshold were locked.

V11. What does `class_weight='balanced'` do conceptually?

It reweights the loss so that errors on the minority class count more, counteracting the class imbalance seen in both datasets (37% and 55% positive prevalence).

V12. Why is a model explanation not a causal explanation?

Feature importance measures predictive dependence within the fitted model, not a mechanism in the world; e.g., worst perimeter's dominance in the Breast Cancer tree reflects a strong statistical association in this sample, not evidence that perimeter causes malignancy.

V13. What is target leakage, and how was it checked?

When a predictor is derived from or otherwise reveals the label. Explicit assertions (`assert 'target' not in X.columns`, `assert 'Malignant' not in X.columns`) confirmed no leakage for the Breast Cancer pipeline.

V14. Why is this laboratory's model not a clinical diagnostic system?

It was trained and evaluated only on small public benchmark datasets without external validation, regulatory review, or prospective clinical testing, and is explicitly documented as educational-use only (Section 17, 23).

20. Common Mistakes Avoided

Potential mistake	How this study avoided it
Calling the model a diagnosis system	Referred to throughout as an educational prediction model / research prototype
Reversing the positive class	Positive class stated and printed explicitly before any metric (1 = Malignant; 1 = Disease present)
Reporting accuracy only	Sensitivity, specificity, precision, F1, balanced accuracy, ROC-AUC, and PR-AUC all reported
Tuning on the test set	GridSearchCV and ccp_alpha search used training-only 5-fold CV; test set scored once
Using predicted labels for ROC-AUC	predict_proba probabilities used for all ROC-AUC / PR-AUC computations
Fitting preprocessing before CV	Heart Disease preprocessing wrapped in a Pipeline/ColumnTransformer inside cross_validate
Displaying an unreadable full tree	Only the top three levels or the compact pruned tree were plotted (Figures 4, 5, 14, 15, 22)
Treating feature importance as causation	Explicitly described as predictive dependence, not causal effect (Sections 9.6, 16.6, 18)
Changing the model after test evaluation	Final test metrics reported as computed on the first and only evaluation pass

21. Submission Guidelines

21.1 File naming

- 23MID0121_Lab02_DecisionTree.ipynb
- 23MID0121_Lab02_Report.pdf / .docx
- 23MID0121_Lab02_CV_Results.csv
- 23MID0121_Lab02_Test_Metrics.csv
- 23MID0121_Lab02_Model.joblib
- 23MID0121_Lab02_ModelCard.md

21.2 Notebook status

All 70 cells were executed from the imports cell through the Model Card cell; outputs are retained and were used verbatim to populate every table and figure in this report.

22. Assessment Rubric — Self-Assessment

Component	Marks	Evidence produced in this study
Problem framing and intended use	8	Positive class, population, and prohibited use stated for both datasets (Section 4, 17)
Dataset provenance and understanding	10	Source, sample unit, class balance documented (Section 8.1, 16.1)
Preprocessing and leakage control	12	Leakage assertions passed; Pipeline/ColumnTransformer used for Heart Disease
EDA and class-imbalance reasoning	8	Class distribution, histograms, correlation heatmap interpreted (Section 11)

Component	Marks	Evidence produced in this study
Decision Tree theory and implementation	14	Gini-based CART implemented and explained (Section 9, 11)
Pruning, tuning, and experimental design	12	GridSearchCV + cost-complexity pruning on training-only CV for both datasets
Evaluation, threshold, and error analysis	16	Full metric suite, threshold search, confusion matrices (Section 13, 16.3)
Interpretation and responsible medical AI	8	Rules, importance, and non-causal language throughout (Section 16.6, 18)
Reproducibility and code quality	6	RANDOM_STATE=42 fixed; joblib artifact saving demonstrated
Industry-style report and presentation	6	This document (Sections 17, 23–24)
Total	100	

23. Responsible Medical AI and Academic Integrity

23.1 Responsible-use statements (as recorded in the notebook Model Card)

Intended use: educational demonstration of Decision Tree and Random Forest algorithms for binary medical classification tasks, and comparison of tree-based learning methods and preprocessing strategies on benchmark medical datasets.

Out-of-scope use:

- This model should not be used for clinical diagnosis.
- It should not replace healthcare professionals.
- It has not been externally validated on real hospital data.
- Predictions must not be interpreted as medical advice.

Limitations (as recorded):

- The datasets are relatively small.
- External validation was not performed.
- Class distributions may differ from real clinical populations.
- Performance may decrease on unseen hospital data.
- Decision Trees are sensitive to the training data, although pruning reduces overfitting.
- Random Forest models are less interpretable than a single Decision Tree.

Fairness and ethical considerations (as recorded):

- The datasets may not represent all patient populations.
- Model predictions could vary across demographic groups.
- Fairness metrics were not explicitly evaluated.
- Human oversight is required before any healthcare decision.

23.2 Academic-integrity statement

All code, tables, and figures in this report are reproduced directly from the executed notebook 23MID0121_Lab02_DecisionTree.ipynb submitted alongside it. No metric or figure has been fabricated or estimated.

24. Final Submission Checklist

- ☑ Positive-class coding stated explicitly and consistently (malignant = 1; disease-present = 1)
- ☑ Dataset source, observation unit, and class balance documented for both datasets
- ☑ Final test set locked before model selection; scored once per dataset
- ☑ Preprocessing (imputation, encoding) wrapped in a Pipeline for the mixed-type Heart Disease data
- ☑ Dummy baseline, Basic CART, Tuned & Pruned CART, and Random Forest results included for both datasets
- ☑ Training-only 5-fold cross-validation used for all model and pruning selection
- ☑ Selected threshold (0.85 / 0.095) justified against a stated 0.90-sensitivity target
- ☑ Final test evaluation includes TN, FP, FN, TP for both datasets
- ☑ Sensitivity, specificity, precision, F1, balanced accuracy, ROC-AUC, PR-AUC reported
- ☑ Tree depth, leaf, and node counts reported before and after pruning
- ☑ Feature importance interpreted predictively, not causally
- ☑ One robustness experiment (5-seed stability, Section 16.5) completed

- ☑ Limitations, fairness constraints, and invalid uses stated (Section 23)
- ☑ Model card and reproducibility details included (Section 17, 23)

Gihub repository link: <https://github.com/Mohan2303-ops/decition-tree.git>

25. References and Dataset Sources

1. Breiman, L., Friedman, J. H., Olshen, R. A., and Stone, C. J. Classification and Regression Trees. Wadsworth, 1984.
2. Hastie, T., Tibshirani, R., and Friedman, J. The Elements of Statistical Learning, 2nd ed. Springer, 2009.
3. James, G., Witten, D., Hastie, T., Tibshirani, R., and Taylor, J. An Introduction to Statistical Learning with Applications in Python. Springer, 2023.
4. scikit-learn developers. Decision Trees and DecisionTreeClassifier documentation, stable release. <https://scikit-learn.org/stable/modules/tree.html>
5. scikit-learn developers. Post-pruning Decision Trees with Cost-Complexity Pruning. https://scikit-learn.org/stable/auto_examples/tree/plot_cost_complexity_pruning.html
6. Wolberg, W., Mangasarian, O., Street, N., and Street, W. Breast Cancer Wisconsin (Diagnostic) [Dataset]. UCI Machine Learning Repository. DOI: 10.24432/C5DW2B.
7. Janosi, A., Steinbrunn, W., Pfisterer, M., and Detrano, R. Heart Disease [Dataset]. UCI Machine Learning Repository. DOI: 10.24432/C52P4X.
8. World Health Organization. Ethics and Governance of Artificial Intelligence for Health: WHO Guidance. Geneva: WHO, 2021.
9. Collins, G. S., et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*, 2024;385:e078378. DOI: 10.1136/bmj-2023-078378.
10. Mitchell, M., et al. Model Cards for Model Reporting. Proceedings of FAT*, 2019.